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EXPT.NO - 5

TITLE - SUM OF SUBSET PROBLEM

In [5]:
```
def Sum_of_Subset(S, d):
    result = []

    def backtrack(i, current_subset, current_sum) :
        if current_sum == d:
            result.append(current_subset)
            return
        if i == len(S) or current_sum > d:
            return
        backtrack(i + 1, current_subset + [S[i]], current_sum + S[i])
        backtrack(i + 1, current_subset , current_sum)

    backtrack(0, [], 0)
    return result

# Input from the user
S = list(map(int,input("ENTER THE ELEMNTS OF SET S:").split()))
d = int(input("Enter the sum d: "))

subsets = Sum_of_Subset(S, d)
print("==================================")
if subsets:
    print("SUBSETS WHOSSE SUM IS : ", d, ":")
    for subset in subsets:
        print(subset)
else:
    print("NO SUBSETS FOUND WITH SUM", d)
print("========================================")
```

```
ENTER THE ELEMNTS OF SET S:1  2 8 4 4 4 5 4 5
Enter the sum d: 10
==================================
SUBSETS WHOSSE SUM IS :  10 :
[1, 4, 5]
[1, 4, 5]
[1, 4, 5]
[1, 4, 5]
[1, 4, 5]
[1, 4, 5]
[1, 5, 4]
[1, 4, 5]
[2, 8]
[2, 4, 4]
[2, 4, 4]
[2, 4, 4]
[2, 4, 4]
[2, 4, 4]
[2, 4, 4]
[5, 5]
========================================
```

In [ ]: